

General Chemistry 2 – Final Exam (Winter 2023)

Chem120 Sections: 001/002

21st April 2023; 2:00 pm to 5:00 pm

Examiner: Prof. Pallavi Sirjoosingh

Assoc. Examiner: Prof. Samuel Sewall

Instructions:

- The exam is a closed-book exam.
- Calculators (non-programmable) are permitted. Crib sheets are not permitted.
- Hand in THIS ENTIRE EXAM BOOKLET
- This booklet has **6 pages (double-sided)** including the cover page.
- Please write legibly and dark enough for a scanner to read. You may use pen or pencil
- **Answer ONLY on the sheets that have a QR code.** Any other sheets will be discarded.
- **Do not write on the QR codes** at the top of the page or your page will not be graded.
- **Do not write beyond the margins and the diagonal line in the top corners.**
- This examination is PRINTED ON BOTH SIDES of the paper
- If you have a question or concern about the exam, please write it on the cover sheet. All questions will be addressed *after* the exam

For all calculations – show your work including equations used. You do not need to show units in the calculations but make sure to show your final answer in the correct units.

Q1 (8 points) Show complete calculations and formula used.

A. (4 points; 3 sig. figs.) Calculate the temperature (in °C) at which the u_{rms} of oxygen gas, O_2 (at 100 °C) is equal to the u_{rms} of chlorine gas, Cl_2 .

B. (4 points; 3 sig. figs.) A sample of sulfur hexafluoride gas occupies 10.21 L at 243°C. Assuming that the pressure remains constant, what temperature (in °C) is needed to reduce the volume to 4.12 L?

Q2 (8 points) Show complete calculations and formula used. Provide final units for your answer.

A. (4 points; 3 sig. figs) Archaeologists use ^{14}C isotope decay to determine the age of an artifact.

^{14}C decay is a first-order process. If 1.44% of the original amount of ^{14}C is present in a wooden tool at the time of analysis, what is the age (in years) of the tool? The half-life of ^{14}C is 5730 years.

B. (4 points; 3 sig figs) A particular reaction has a rate constant, $k = 2.12 \times 10^{-2} \text{ M}^{-1}\text{min}^{-1}$. If initial concentration of the reactant is 0.135 M, how much reactant remains after 3.5 hours? What is the rate of the reaction initially?

Q3 (8 points)

A 1.431 kg sample of N_2 gas is at equilibrium in a piston chamber under an external pressure of 11.9 atm at $T=298.15$ K. The external pressure is very rapidly increased to 25.0 atm and the temperature is held constant.

A. (5 points; 3 sig figs) Calculate the pressure-volume work (in J) for this change and indicate if it is an expansion or a compression.

B. (3 points) Sketch the process on a pressure versus volume plot labeling key points (initial pressure, initial volume, final pressure, final volume) on the plot, and depict the work magnitude.

Q4 (10 points)

Consider the reaction below:

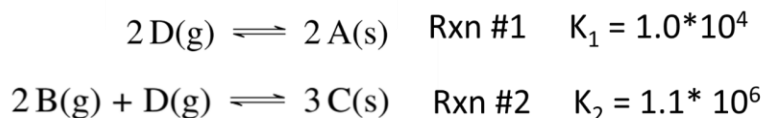


The equilibrium constant was measured at a temperature of 298 K, using units of Molarity.

A. (5 points; 3 sig. figs.) Consider a system prepared by adding 0.25 moles of A into a 1.00 Liter container. Calculate the concentrations of A and B when the system reaches equilibrium. Note that neither concentration can be zero at equilibrium

B. (5 points; 3 sig. figs.) Consider a system prepared by adding 2 moles of B into a 1.00 Liter container. Calculate the concentrations of A and B when the system reaches equilibrium. Note that neither concentration can be zero at equilibrium

Q5 (9 points)



Consider the two reactions, Rxn #1 and #2, above. The equilibrium constants given are for the gas pressures measured in Bar, for both reactions these values were determined at 575 K.

A. (3 points) Looking at reaction 1, predict the signs (positive or negative) of ΔG_r^0 , ΔH_r^0 , and ΔS_r^0 . Briefly explain your reasoning for each prediction.

B. (4 points) You have been tasked to design a catalyst that can enhance the interconversion of A, B, C and D in the gas phase. Using any manipulations necessary on reactions 1 and 2 determine the equilibrium constant (K_3) of reaction 3:



C. (2 points) Should you increase or decrease the temperature if you want to have the equilibrium farther towards the products for reaction 3? Explain your answer.

Q6 (7 points)

A. (4 points) A new element has been discovered, Sewallium (Sw)! It forms an ionic compound with chlorine that acts as a strong electrolyte in water. You are in charge of determining the number of chloride ions bound to each Sewallium atom, using colligative properties.

In a carefully controlled experiment a 0.002 molal solution of Sewallium Chloride in water has a boiling point elevation of 0.0051 K. For water, $K_b = 0.512\text{ }^{\circ}\text{C}/(\text{mol}/\text{kg})$.

Using this measurement determine the most likely chemical formula for Sewallium Chloride from among the following options: SwCl, SwCl₂, SwCl₃, SwCl₄. Show your work.

B. (3 points) What is the solubility of Ag₂SO₄ at 25 °C? Give your answer in g/mL. (K_{sp} of Ag₂SO₄ is 1.20×10^{-5}). Show your work.

Q7 (10 points)

Using the data below, answer the following questions (Assume $T = 298\text{ K}$)

Compound	pK_a
HCOOH	4.75
H ₃ PO ₄	2.12
H ₂ PO ₄ ⁻	7.21

A. (2 points)

Among the given compounds, which is the weakest acid? Among the conjugates of the given compounds, which is the weakest base? Explain your reasoning

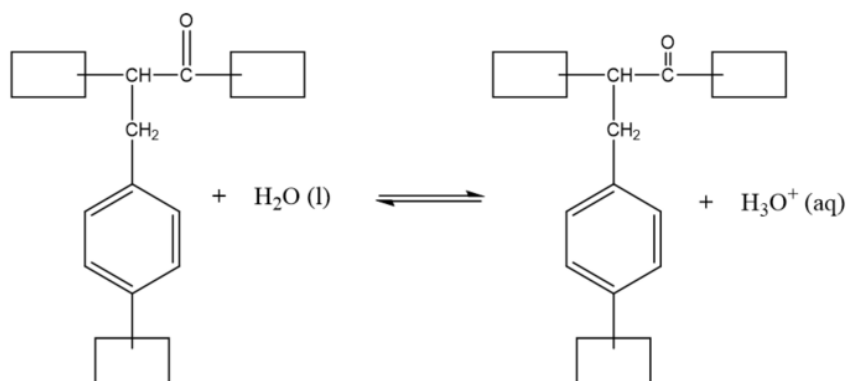
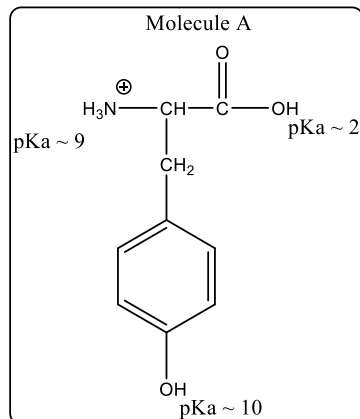
B. (3 points)

Which acid/conjugate base combination would be the most appropriate buffer system for performing an experiment at $\text{pH} = 5.10$. Explain your reasoning.

Q8 (10 points)

A. (3 pts) The three acidic hydrogens and their pK_a s are marked for the Molecule A (left box). Complete the chemical equation provided on the right ($pH = 3.00$).

In the boxes provided, write down the three groups ($-OH$; $pK_a = 2$, $-NH_3^+$; $pK_a = 9$, and $-OH$; $pK_a = 10$) in their appropriate acidic or conjugate basic forms, for the chemical equation representing the equilibria between the two most prominent states at $pH = 3.0$.



B. (7 pts; 3 sig figs) Two buffer systems ($pH = 6.38$) are prepared:

Buffer A: 100 mL total of 0.1500 M Na_2CO_3 and 0.1500 M $NaHCO_3$

Buffer B: 100 mL total of 0.0150 M Na_2CO_3 and 0.0150 M $NaHCO_3$.

Which of the two would be a better choice for a buffer? Justify your choice by **showing calculations** for the change in pH (for both buffers A and B) when 50.00 mL of 0.01 M $NaOH$ is added to the system.

Q9) 10 points (Assume $T = 298\text{ K}$)

60.00 mL of a 0.2500 M solution of a weak acid ($pK_a = 7.21$) is titrated with 0.2000 M strong base, NaOH.

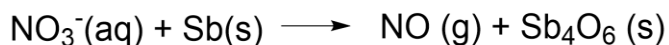
A. (4 points; 4 sig figs) Calculate the pH after addition of 10.00 mL of NaOH. Show your calculations

B. (4 points; 4 sig figs) Calculate the pH at equivalence point. Show your calculations.

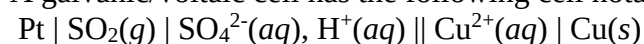
C. (2 points) At what volume (mL) of NaOH added, will the $\text{pH} = 7.21$. Briefly explain your answer.

Q10) 10 points

A. (4 points) Balance the following equation (in acidic media). Show each half-reaction and the fully balanced equation using smallest whole number coefficients.

**B) (4 points)**

A galvanic/voltaic cell has the following cell notation.



Write down the reaction occurring at the anode, the cathode, and correctly balanced spontaneous reaction. (Pt is an inactive electrode)

C) (2 pts) Using the table given below, answer the following questions:

Half reactions	E° (volts)
$\text{A}^+(\text{aq}) + \text{e}^- \rightarrow \text{A}(\text{s})$	-1.31
$\text{B}^+(\text{aq}) + \text{e}^- \rightarrow \text{B}(\text{s})$	2.01
$\text{C}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{C}(\text{s})$	1.43
$\text{X}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{X}^-(\text{aq})$	-0.55

(i) Which reagent is the strongest reducing agent?

(ii) Which reagent(s), if any, will oxidize C(s)?

Q11) 10 points

A) (6 points; 3 sig. figs.) A voltaic cell with Ni/Ni^{2+} ($E^\circ = -0.25 \text{ V}$) and Co/Co^{2+} ($E^\circ = -0.28 \text{ V}$) half-cells has the following initial concentrations: $[\text{Ni}^{2+}]_{\text{in}} = 0.80 \text{ mol/L}$ and $[\text{Co}^{2+}]_{\text{in}} = 0.20 \text{ mol/L}$.

(i) What is the initial E_{cell} ? Show your work.

(ii) Will the concentration of Ni^{2+} ions at equilibrium ($[\text{Ni}^{2+}]_{\text{eq}}$) be greater than, equal to, or less than the concentration of Ni^{2+} ions initially ($[\text{Ni}^{2+}]_{\text{in}}$)? Explain your answer.

B) (4 points; 3 sig. figs.) How much time (s) does it take to deposit 82.5 g of Zn on a steel gate when 21.0 A is passed through a ZnSO_4 solution?